TICKET BOOKING SYSTEM DESIGN

{MOVIE}

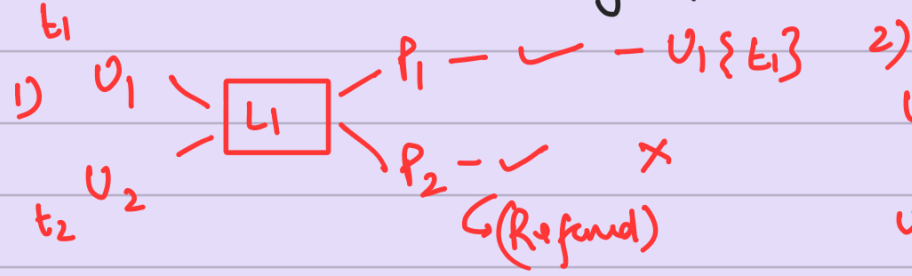
BOOK MY SHOW	DISTRICT

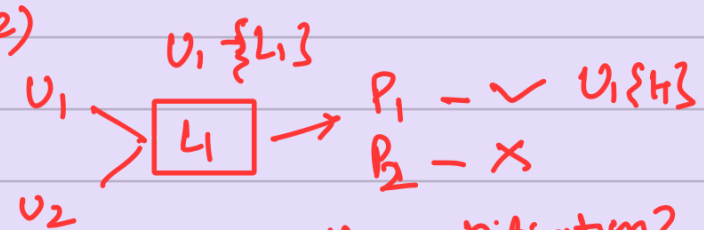
Requirements

(*) 1) Book the ticket
(-) 2) Search movie, view Details
(-) 3) Search All the cinema (PVR) near you.
(*) 4) Highly Concurrent
(x) 5) Payment (not going in Details)
(x) 6) Comment (not going in Details)
(x) 7) Send ticket (notification → (not going in Detail)

Approach :-

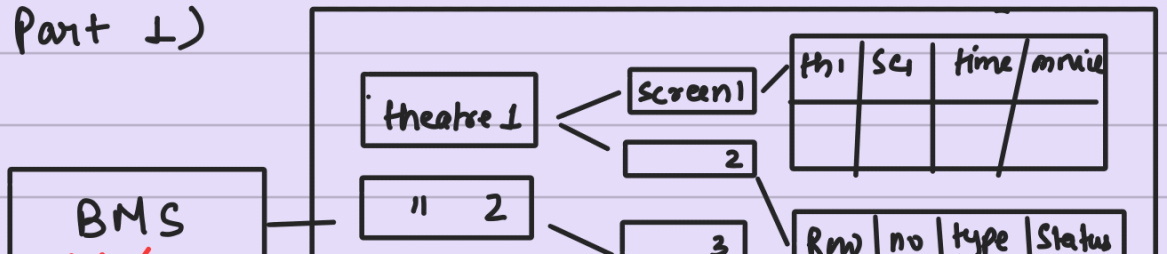
1) ticket claim after payment confirmed.
2) ticket claim with some locking strategy.
(*) the person who come first will get the lock.

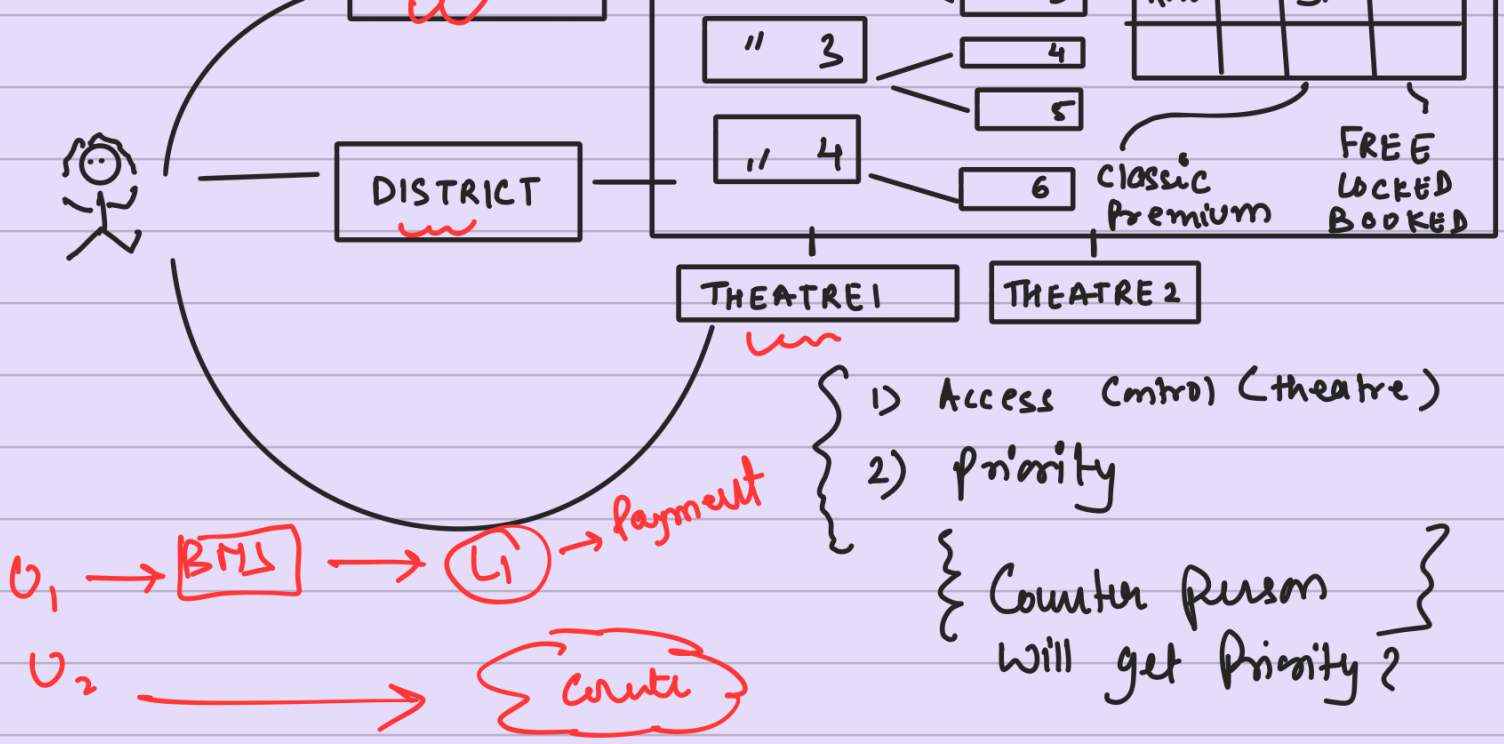




→ U2 notification?
→ U1 x, U2, U1 ✓

Concept 1 (Book the ticket Part 1)





1) Access Control (theatre)
2) Priority

Counter person will get Priority ?

U_1 → BMS → L1 → Payment

U_2 → counter

Concept 2 (Understanding DB Relation)

1) All the above table {Relational DB}
2) movie → { mango DB }
   → Document type Data. → Raw info

{
name, -
actors - [ ]
Co-actor list [, , ,]
ratings : [imdb : __
BMS : __]
active -
}

3) Mapping table

theatreid	screenid	time	movie

Concept 3 (Search movie all the cinema, near you)

1) Device {Lat, Lng}

2) Ask where do you want to Book ticket.

1) → search on the basis of city

2) → Search on the basis of coordinate · distance.

3) → By name search

{ fuzzy search
soundex search }

Concept 4 :- (Highly Concurrency)

{ Concurrency Control }

Isolation level	Dirty Read	Non-Repeatable Read	phantom Read	Control
Read uncommit	Yes	Yes	Yes	optimistic ↑
Read commit	No	Yes	Yes	optimistic ↓
Repeatable Read	No	No	Yes	pessimistic ↓
Serializable	No	No	No	pessimistic ↑

high

low

1) which one do you need ?

1) pessimistic

Pro	Cons
1) Prevent double booking	1) Deadlock
2) lock the seat During booking {Row level locking}	2) Slow

Concept 5 :- { UI }

thid	seatid	Seatid	Status	Userid	Starttime	expiry
✓	✓	123	Free	null	null	null
			Locked	123	x	(x+10)

(Status = free X)

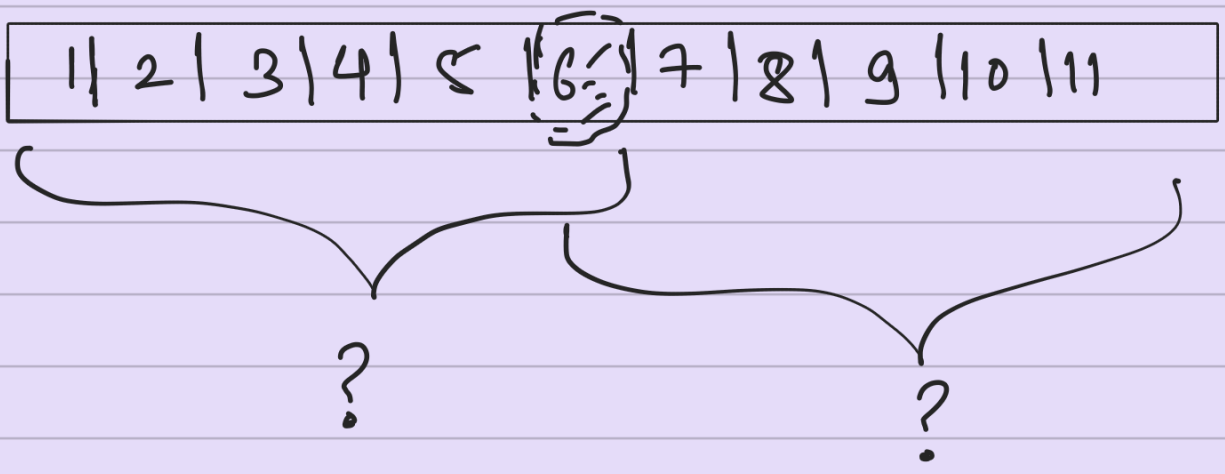
Status != Booked and time > expirytime { free }

v1 - locked - (x - x+10) (x+12)

(write)

Stored proc, Redis, DB trigger
Cron.

Concept - 6



→ I had to rollback.

→ should we gave them middle ground seats. of booking available

final Design